Ricky Tang

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Education

University of Waterloo — GPA: 3.9/4.0
Honours Bachelor of Software Engineering (Co-op) — *Awarded $11,000+ in scholarships*

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Go, C#, C++, Bash, Swift, C, HTML, CSS
Technologies: React, Next.js, FastAPI, Node.js, Express, ASP.NET Core, Flask, RESTful APIs, GraphQL, WebSockets, PyTorch
Tools: Git, Docker, Linux, CI/CD, Redis, MySQL, PostgreSQL, SQLite, MongoDB, ChromaDB, Supabase, AWS, GCP, Prometheus

Experience

Hamming AI (YC S24) — May 2026 – Present
Software Engineer — *San Francisco, CA*
- Building automated infra for testing AI voice agents at scale, generating **1000+** parallel calls/minute with **50+** metrics

WAT.ai | *Python, FastAPI, LangChain, ChromaDB, TypeScript, React, DataLab* — Jan. 2026 – Present
Software Engineer — *Waterloo, ON*
- Developed self-correcting RAG system for **174-page EPA regulatory PDFs** in collaboration with **Bindwell (YC W25)**
- Achieved **1.00 faithfulness score** (zero hallucinations) via **DeepEval** benchmarking across golden evaluation dataset
- Engineered recursive table restoration with **100% fragment recovery** and **26% token reduction** via uniqueness filtering
- Implemented hybrid **vector + BM25** retrieval with lazy sampling, achieving **20-100x speedup** over full concatenation

WATonomous | *C++, ROS 2, Docker* — Jan. 2026 – Present
Software Engineer — *Waterloo, ON*
- Architected complete **ROS 2 navigation stack** including Costmap, Map Memory, A* Planner, and Pure Pursuit Controller
- Engineered **adaptive A* planner** with iterative cost relaxation, achieving **100% path success** in constrained spaces
- Amplified grid resolution by **100% (5cm)** with a **50% larger safety buffer (1.5m)**, eliminating corner-cutting collisions

Waterloo Aerial Robotics Group (WARG) | *Python, React, Flask, OpenCV, MAVLink, Docker* — Oct. 2025 – Present
Software Engineer — *Waterloo, ON*
- Streamlined ground station UI with one-click pause/resume for missions, eliminating manual switching for **50+ operators**
- Reduced mission failure recovery time from 30s+ to 3-5s for command pipeline operations (**85% improvement**)
- Engineered full-stack control pipeline with React frontend, Flask-SocketIO backend, and MAVLink for real-time commands
- Implemented OpenCV object detection in aerial imagery and MAVLink telemetry streaming, achieving **80%+ IoU accuracy**

Projects

Tark – Google Earth for Game Devs | *TypeScript, Next.js, Python, FastAPI, Redis, Prometheus, Leaflet, SciPy, Numpy, Pytest*
- Developed web app turning locations into game-ready 3D meshes in **<15 seconds** (typically weeks of manual modeling)
- Processed Mapbox elevation and satellite imagery to generate terrain meshes at **45K+ triangulated faces per second**
- Extracted **2000+ building footprints** from OpenStreetMap and generated textured .obj files for Unity/Blender workflows
- Productionized with **Redis** job store, reducing latency by **30%** via asyncio workers with **98.2% accuracy** in pytest suites

BrainLattice – PDF to Obsidian Vault Generator | *Python, FastAPI, PostgreSQL, Redis, TypeScript, React, AWS, Cloudflare*
- Developed React + FastAPI web app and **npm CLI** converting PDFs into Obsidian vaults with NetworkX graph generation
- Reduced token costs by **70-80%** via global concept seeds and batched extraction replacing full-document context caching
- Accelerated processing **70%** via parallelization and embedding deduplication, generating **100+ node graphs** in **<1 minute**
- Built event-driven pipeline using **QStash + Lambda + R2** with **pgvector RAG** for on-demand notes, reaching **300+ installs**

Quota – VS Code Extension for Cost Optimization | *TypeScript, VS Code API, LangChain, Python, FastAPI, Next.js, MongoDB, GCP*
- Developed VS Code extension analyzing API/cloud/DB overhead with inline annotations and optimization suggestions
- Achieved <7s initial indexing and **<1s** refresh analysis via file hash caching, outperforming AI IDEs by **47x in speed**
- Implemented hybrid AST + regex parser detecting **2x more cost issues** than AI IDEs while saving **~50k tokens per check**
- Architected RAG-powered web sandbox with **LangChain + FAISS** delivering architectural recommendations in **<2s**

Awards

- DeltaHacks 2026: **1st Place Overall** | Hack Western 2025: **Best AI Application Built with Cloudflare**
- GenAI Genesis 2026 (Canada's largest AI hackathon): **Top 10 Overall** (out of 250+ teams)